SOUMMA CHAKRABORTY

MECHANICAL ENGINEER // CAD/CAM & PRECISION CNC MACHINE DESIGN

Email: soummachakraborty.eng@gmail.com | Location: India (Open to Global Relocation)
Portfolio: https://soumma-chakraborty.vercel.app
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1. PROFESSIONAL SUMMARY

Mechanical Engineer specializing in CNC machine architecture, precision kinematics, CAD/CAM integration, and structural FEA simulation. Experienced in translating product concepts into high-rigidity assemblies, sheet metal stamping press tools, plastic injection moulds, and ASME Y14.5 / ISO 2768-m production drawings.

2. TECHNICAL COMPETENCIES & TOOLS MATRIX

CAD & 3D Modeling: SolidWorks, Siemens NX, PTC Creo Parametric, Autodesk Fusion 360, AutoCAD (2D Drafting)
CAM & CNC Machining: Delcam PowerShape, Delcam PowerMill, Mastercam, Multi-Axis Toolpaths, Wire EDM, G/M-Code
Tool & Die Design: Sheet Metal Press Tools, Blanking/Piercing, Plastic Injection Moulds, Strip Layouts, Core/Cavity
Analysis & Standards: ANSYS FEA (Structural & Thermal Deflection), GD&T (ASME Y14.5 / ISO 1101), ISO 2768-m, DFM/DFA
Quality & Metrology: 7 QC Tools, Vernier/Micrometers, Dial Indicators (TIR), Surface Roughness (Ra/Rz), CMM Principles

3. PROFESSIONAL EXPERIENCE & INDUSTRY TRAINING

Quality Assurance & Mechanical Engineering Associate

Mechanical Tooling & Manufacturing Environment | *India*

* Executed dimensional inspection and runout tolerances (TIR) according to ASME Y14.5 and ISO 2768-m.
* Applied 7 QC Tools (Pareto Analysis, Cause & Effect Diagrams, Control Charts) to resolve manufacturing variances.
* Collaborated with CNC machinists and toolroom engineers to optimize milling, turning, and EDM workflows.

4. FEATURED ENGINEERING & TOOLING PROJECTS

Project 1: 'KEY RING' Press Tool Design, Manufacture & Stamping Tryout

SolidWorks CAD | *AutoCAD Strip Layout* | *Mastercam CAM* | *Wire EDM & CNC Machining* | *Toolroom Assembly*

* Modeled 3D product in SolidWorks; designed progressive strip layout in AutoCAD for maximum material yield.
* Generated multi-axis CNC toolpaths with Mastercam; fabricated hardened punch/die plates using Wire EDM & CNC.
* Assembled punch plate, stripper, and guide pillars; stamped out defect-free production key rings in trial press run.

Project 2: Hand Injection Mould Machine Design through Reverse Engineering

Reverse Engineering | *SolidWorks 3D Assembly* | *Precision CNC Machining* | *Full Mechanical Integration*

* Measured reference machine parts using precision micrometers and calipers to determine critical baseline dimensions.
* Built complete kinematic 3D CAD assembly in SolidWorks, verifying slide clearances and tolerance fits (H7/g6).
* Fabricated plunger mechanism, barrel, and clamping frame on CNC centers; successfully assembled complete machine.

Project 3: Drone Prototype Design & Multi-Axis Manufacturing

Delcam PowerShape Surfacing | *Delcam PowerMill CAM* | *CNC Machining & Wire EDM* | *Aircraft Aluminum 6061-T6*

* Modeled lightweight aerodynamic curves and structural arm mounts using Delcam PowerShape 3D surfacing.
* Programmed high-speed milling strategies in Delcam PowerMill for flawless surface finish and rapid cycle times.
* Fabricated intricate frame arms and motor mount brackets with zero thermal distortion via Wire EDM and CNC.

SOUMMA CHAKRABORTY -- CURRICULUM VITAE (CONT.)

Mechanical Engineer | CNC Machine Design, CAD/CAM, GD&T & FEA Simulation

FEATURED PROJECTS (CONTINUED)

Project 4: 'MARKER STAND' Plastic Injection Mould Design, Simulation & Production

SolidWorks Parting Surfaces | AutoCAD Stackup | ANSYS FEA Validation | Mould Machine Tryout

* Designed Marker Stand product and extracted core and cavity blocks with proper draft angles and shrinkage factors.
* Structurally verified mould plate deflection under continuous injection pressure using ANSYS FEA simulation.
* Machined mould plates, guide pillars, and cooling lines via Mastercam and CNC; produced high-finish parts.

5. ACADEMIC & TECHNICAL QUALIFICATIONS

Post Graduate Diploma in Tool Design & CAD/CAM (PGDTD&CC)

Central Tool Room & Training Centre (CTTC), Bhubaneswar | Ministry of MSME, Govt. of India

* Specialized in Press Tool Design, Plastic Injection Mould Design, Die Casting, and Multi-Axis CAM Programming.
* Rigorous hands-on training on CNC Machining Centers, CNC Turning, Wire EDM, and Precision Surface Grinding.

Bachelor of Technology (B.Tech) in Mechanical Engineering

Undergraduate Engineering Degree | India

* Core focus: Theory of Machines, Machine Design, Fluid Mechanics, Thermodynamics, Strength of Materials, Metrology.
* Applied coursework in CAD modeling, GD&T geometric tolerancing, kinematics, and manufacturing technology.

6. VERIFIED CERTIFICATIONS & ACCREDITATIONS

CTTC Bhubaneswar (Govt. of India MSME Center)

*** Post Graduate Diploma in Tool Design & CAD/CAM:**
SolidWorks, Siemens NX, PTC Creo Parametric, Autodesk Fusion 360

*** Advanced CAD/CAM Technologies:**
Delcam PowerShape, Delcam PowerMill, Mastercam Milling & Turning

*** Industrial CAM Toolpath Synthesis:**
Accreditations from IIT Roorkee, IIT Kanpur, and IIT Kharagpur

*** IIT / NPTEL Engineering Certifications:**
GD&T (ASME Y14.5 / ISO 1101), ISO 2768-m Precision Standards, 7 QC Tools

*** Engineering Standards & Quality:**
ANSYS FEA Static Structural & Deflection Simulation

*** Structural & Thermal Simulation:**

7. PROFESSIONAL REFERENCES & DIGITAL VERIFICATION

Interactive Engineering Portfolio: https://soumma-chakraborty.vercel.app

LinkedIn Professional Profile: https://www.linkedin.com/in/soumma-chakraborty-10b85a230/

Direct Contact / Inquiries: soummachakraborty.eng@gmail.com

Standard Drawing Compliance: ASME Y14.5M, ISO 2768-m, DIN 7168, ISO 1101